

101. If three successive terms of a G.P. with common ratio $r(r > 1)$ are the lengths of the sides of a triangle and $[r]$ denotes the greatest integer less than or equal to r , then $3[r] + [-r]$ is equal to _____.

यदि सार्व अनुपात $r(r > 1)$ की एक G. P. के तीन क्रमागत पद एक त्रिभुज की भुजाओं की लंबाईयों हैं तथा $[r]$ महत्तम पूर्णांक $\leq r$ है, तो $3[r] + [-r]$ बराबर है _____

Ans. 1

Sol. Let three terms of GP are $\frac{a}{r}, a, ar(r > 1)$

Sum of two smaller sides $>$ third side

$$\Rightarrow \frac{a}{r} + a > ar$$

$$\Rightarrow 1 + r > r^2$$

$$\Rightarrow r^2 - r - 1 < 0$$

$$\Rightarrow \frac{1-\sqrt{5}}{2} < r < \frac{1+\sqrt{5}}{2}$$

$$\text{but } r > 1 \Rightarrow r \in \left(1, \frac{1+\sqrt{5}}{2}\right)$$

$$\Rightarrow [r] = 1 \text{ and } [-r] = -2$$

$$\text{So } 3[r] + [-r] = 3 - 2 = 1$$

102. If a and b are the roots of the equation $x^2 - 7x - 1 = 0$, then the value of $\frac{a^{21} + b^{21} + a^{17} + b^{17}}{a^{19} + b^{19}}$ is equal to _____.

यदि समीकरण $x^2 - 7x - 1 = 0$ के मूल a तथा b हैं, तो $\frac{a^{21} + b^{21} + a^{17} + b^{17}}{a^{19} + b^{19}}$ का मान बराबर है।

Ans. 51

Sol. $x^2 - 7x - 1 = 0$ $\leq_a$

By newton's theorem

$$S_{n+2} - 7S_{n+1} - S_n = 0$$

$$S_{21} - 7S_{20} - S_{19} = 0$$

$$S_{20} - 7S_{19} - S_{18} = 0$$

$$S_{19} - 7S_{18} - S_{17} = 0$$

$$\begin{aligned}
 \frac{S_{21} + S_{17}}{S_{19}} &= \frac{S_{21} + (S_{19} - 7S_{18})}{S_{19}} \\
 &= \frac{S_{21} + S_{19} - 7(S_{20} - 7S_{19})}{S_{19}} \\
 &= \frac{50S_{19} + (S_{21} - 7S_{20})}{S_{19}} \\
 &= 51 \cdot \frac{S_{19}}{S_{19}} = 51
 \end{aligned}$$

103. Let M denote the set of all real matrices of order 3×3 and let $S = \{-3, -2, -1, 1, 2\}$. Let

$$S_1 = \{A = [a_{ij}] \in M : A = A^T \text{ and } a_{ij} \in S, \forall i, j\},$$

$$S_2 = \{A = [a_{ij}] \in M : A = -A^T \text{ and } a_{ij} \in S, \forall i, j\},$$

$$S_3 = \{A = [a_{ij}] \in M : a_{11} + a_{22} + a_{33} = 0 \text{ and } a_{ij} \in S, \forall i, j\},$$

If $n(S_1 \cup S_2 \cup S_3) = 125\alpha$, then α equals _____.

माना कि M , 3×3 कोटि के सभी वास्तविक आव्यूहों के समुच्चय को निर्दिष्ट करता है और $S = \{-3, -2, -1, 1, 2\}$ है। माना कि

$$S_1 = \{A = [a_{ij}] \in M : A = A^T \text{ और } a_{ij} \in S, \forall i, j\},$$

$$S_2 = \{A = [a_{ij}] \in M : A = -A^T \text{ और } a_{ij} \in S, \forall i, j\},$$

$$S_3 = \{A = [a_{ij}] \in M : a_{11} + a_{22} + a_{33} = 0 \text{ और } a_{ij} \in S, \forall i, j\} \text{ है।}$$

यदि $n(S_1 \cup S_2 \cup S_3) = 125\alpha$ है, तो α बराबर है _____.

Ans. 1613

Sol.
$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

No. of elements in $S_1 : A = A^T \Rightarrow 5^3 \times 5^3$

No. of elements in $A = -A^T \Rightarrow 0$

since no zero in S

$$\text{No. of elements in } S_3 \Rightarrow a_{11} + a_{22} + a_{33} = 0 \Rightarrow \left. \begin{array}{l} (1, 2, -3) \Rightarrow 31 \\ \text{or} \\ (1, 1, -2) \Rightarrow 3 \\ \text{or} \\ (-1, -1, 2) \Rightarrow 3 \end{array} \right\} \Rightarrow 12 \times 5^6$$

$$n(S_1 \cap S_3) = 12 \times 5^3$$

$$n(S_1 \cup S_2 \cup S_3) = 5^6(1 + 12) - 12 \times 5^3$$

$$\Rightarrow 5^3 \times [13 \times 5^3 - 12] = 125\alpha$$

$$\alpha = 1613$$

$$\frac{\alpha}{\beta} = 15$$